

LEARNING PANDAS

I came for cute pandas, I stayed for DataFrames.

WHAT I THOUGHT:



CUTE PANDAS
EATING BAMBOO 🌿

GROUP BY LIFE,
AGGREGATE
THE PROBLEMS!

WHAT I'M ACTUALLY DOING:

```
import pandas as pd

df = pd.read_csv('life.csv')

summary = (
    df.groupby('problem')
      .agg({'stress': 'sum',
            'sleep': 'mean'})
      .reset_index()
)
```

problem	stress	sleep
Exams	999	2.3
Bugs	888	3.1
Deadlines	777	2.0
Life	404	1.5



Note to self:
Don't forget
.dropna()
in real life too 😊



PANDAS: BECAUSE ADULTING IS HARD,
BUT DATA CLEANING HELPS! 🐼

WHAT IS PANDAS? (Python Data Analysis Library)

KEY FEATURES

Name	Age	Score
Name	24	70
Jane	26	80
Nalia	27	50
Score	34	60

Series

DataFrame

Two-dimensional, labeled data structure, like a spreadsheet.



Data Cleaning & Prep

Tools for handling missing data, filtering, and transforming.

Data I/O



Load and save from various formats: CSV, Excel, JSON, SQL.



ANALYSIS TOOLS



Exploration & Aggregation

Statistical analysis, grouping, and aggregating data.

0	0
1	1
2	2

Series

One-dimensional, labeled array of any data type.

Integration



Seamless integration with other Python libraries.

Powerful, flexible data manipulation for Python

WHY USE IT? AND WHERE CAN YOU USE IT? (Pandas Python Data Library)

WHY USE PANDAS?



EFFICIENCY

Process vast amounts of data quickly and easily.



VERSATILITY

Handles diverse data types, complex cleaning tasks, and statistical analysis.



PRODUCTIVITY



Clear syntax and built-in functions speed up the entire data workflow.



WHERE CAN YOU USE IT?



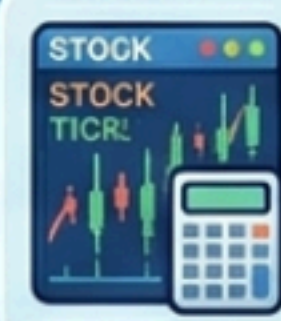
DATA SCIENCE & ML

Essential for data wrangling, feature engineering, and model training in machine learning. Regression Clustering



BUSINESS INTELLIGENCE

Analyze customer data, sales figures, and financial performance for insights.



FINANCE & ECONOMICS

Analyze time-series financial data and economic indicators.



ACADEMIC RESEARCH & SCIENTIFIC DATA

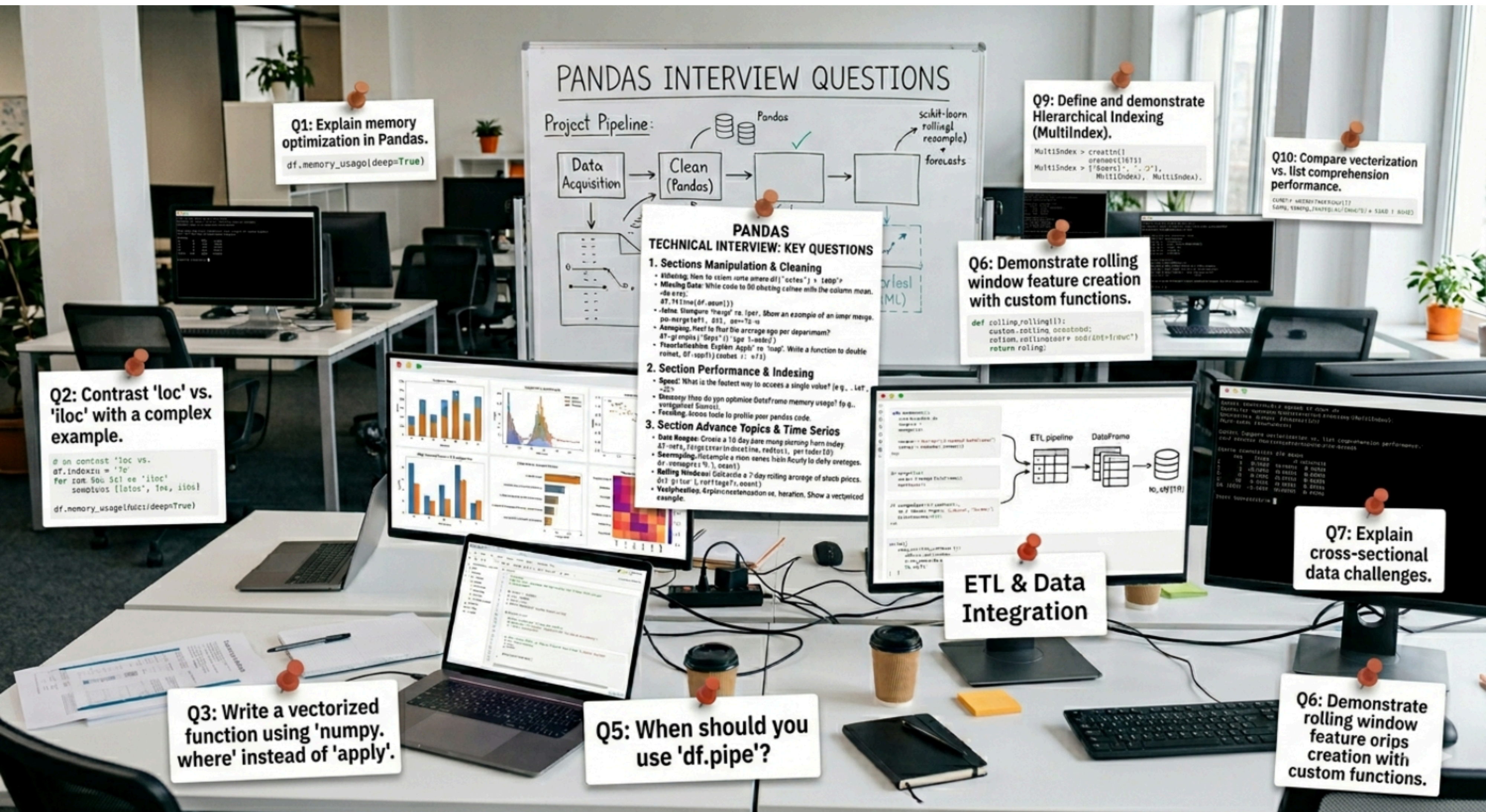
Used across various fields like physics, social sciences, and bioinformatics to analyze data.

PREPARE WITH THE BEST

TOP 20 PANDAS INTERVIEW QUESTIONS



Source: General Community Knowledge



Q1: Explain memory optimization in Pandas.

```
df.memory_usage(deep=True)
```

Q2: Contrast 'loc' vs. 'iloc' with a complex example.

```
@ on context 'loc' vs.
df.index = '7e'
for row, col in df.iterrows():
    df.loc[row, col] = 1
df.memory_usage(deep=True)
```

Q3: Write a vectorized function using 'numpy.where' instead of 'apply'.

Q5: When should you use 'df.pipe'?

Q9: Define and demonstrate Hierarchical Indexing (MultiIndex).

```
MultiIndex = createMultiIndex(
    createMultiIndex(100000,
    MultiIndex = ['Series', 'Series'],
    MultiIndex = MultiIndex)
```

Q10: Compare vectorization vs. list comprehension performance.

```
sum(df['Series'])
sum(df['Series'])
```

Q6: Demonstrate rolling window feature creation with custom functions.

```
def rolling_rolling():
    custom_rolling = lambda df: df.rolling(window=10).sum()
    return rolling_rolling()
```

ETL & Data Integration

Q7: Explain cross-sectional data challenges.

Q6: Demonstrate rolling window feature creation with custom functions.

PANDAS PROJECTS

Hands-on data analysis projects using Python Pandas



1 Sales Analysis

Analyze sales data to find top-selling products, monthly revenue and trends.

Dataset: sales.csv

Key Pandas Functions
groupby(), agg(), sort_values(),
reset_index(), to_datetime()



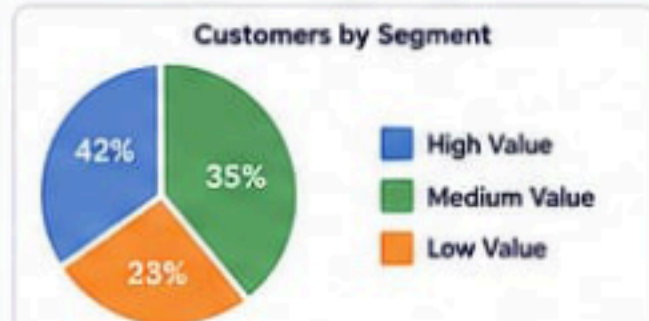
Outcome / Insights
Identify best selling products and monthly revenue trends.

2 Customer Segmentation

Segment customers based on purchase behavior and demographics.

Dataset: customers.csv, orders.csv

Key Pandas Functions
merge(), groupby(), agg(), qcut(),
value_counts(), pivot_table()



Outcome / Insights
Group customers for targeted marketing and better engagement.

3 Data Cleaning & Preprocessing

Clean raw data by handling missing values, outliers and inconsistent formats.

Dataset: raw_data.csv

Key Pandas Functions
isnull(), fillna(), dropna(), replace(),
dtype, astype(), strip(), apply()

Before Cleaning			After Cleaning		
ID	Age	Salary	ID	Age	Salary
1	25	50000	1	25	50000
2	-	60000	2	27	60000
3	30	NaN	3	30	57000
4	28	52000	4	28	52000
5	-	58000	5	32	58000

Outcome / Insights
Clean and structured data ready for analysis.

4 Exploratory Data Analysis (EDA)

Explore data to find patterns, correlations and key statistics.

Dataset: e-commerce.csv

Key Pandas Functions
describe(), info(), value_counts(),
corr(), nunique(), pivot_table()



Outcome / Insights
Understand data distribution, relationships and key drivers.

5 Time Series Analysis

Analyze time-based data to identify trends, seasonality and patterns.

Dataset: stock_prices.csv

Key Pandas Functions
to_datetime(), resample(), rolling(),
shift(), pct_change(), diff()



Outcome / Insights
Identify trends and make future predictions.

6 Top Movies Analysis

Analyze movie data to find top-rated movies, genres and voting patterns.

Dataset: movies.csv

Key Pandas Functions
sort_values(), value_counts(), groupby(),
mean(), idxmax(), nlargest()

Top 5 Movies by Rating			
Movie Title	Genre	Rating	Votes
The Shawshank Redemption	Drama	9.3	2343110
The Godfather	Crime	9.2	1620367
The Dark Knight	Action	9.0	2303332
The Godfather: Part II	Crime	9.0	1129952
12 Angry Men	Drama	9.0	689845

Outcome / Insights
Discover highly rated movies and popular genres.

7 Student Performance Analysis

Analyze student performance based on marks, attendance and subjects.

Dataset: students.csv

Key Pandas Functions
groupby(), agg(), pivot_table(),
mean(), max(), min()

Average Marks by Subject	
Subject	Average Marks
Math	78.4
Science	82.1
English	74.6
History	68.3
Computer	87.5

Outcome / Insights
Identify top-performing subjects and students.

8 Employee Data Analysis

Analyze employee data for salary trends, departments and experience.

Dataset: employees.csv

Key Pandas Functions
groupby(), agg(), value_counts(),
sort_values(), describe()

Average Salary by Department		
Department	Avg Salary	Employees
IT	85000	45
HR	62000	20
Finance	75000	30
Marketing	65000	25
Sales	70000	40

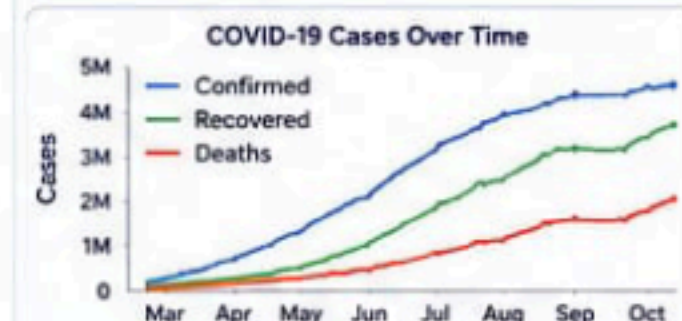
Outcome / Insights
Understand salary distribution and workforce insights.

9 COVID-19 Data Analysis

Analyze COVID-19 data to track cases, recoveries and death rates.

Dataset: covid_19.csv

Key Pandas Functions
groupby(), resample(), sum(), diff(),
rolling(), plot()



Outcome / Insights
Track pandemic progression and recovery rate.

10 Retail Store Analysis

Analyze retail store data to optimize inventory, sales and profits.

Dataset: retail_store.csv

Key Pandas Functions
pivot_table(), groupby(), agg(),
merge(), fillna(), crosstab()

Sales by Region		
Region	Total Sales	Total Profit
North	1,250,000	150,000
South	980,000	120,000
East	1,100,000	130,000
West	1,050,000	125,000

Outcome / Insights
Optimize sales strategy and inventory planning.

Why Pandas?

- ✓ Fast and efficient data analysis
- ✓ Easy data cleaning & manipulation
- ✓ Powerful handling of structured data



Core Pandas Skills Covered

- Data Structures (Series, DataFrame)
- GroupBy Operations
- Sorting & Ranking
- Data Selection & Filtering
- Data Aggregation
- Time Series Analysis
- Handling Missing Data
- Merging & Joining Data
- Data Visualization

Benefits

- ✓ Real-world data analysis experience
- ✓ Stronger portfolio projects
- ✓ Better job opportunities in Data Science



No.	Question	What It Tests / Key Concept
1.	What is Pandas? What are its main data structures?	Basics: Series, DataFrame, Index
2.	How do you create a Pandas Series?	Series creation
3.	How do you create a DataFrame?	DataFrame creation
4.	How do you read data from a CSV file?	I/O – read_csv()
5.	How do you read data from an Excel file?	I/O – read_excel()
6.	How do you display the first and last n rows of a DataFrame?	head(), tail()
7.	How do you get information about a DataFrame?	info(), dtypes, shape
8.	How do you select a single column?	Column selection
9.	How do you select multiple columns?	Multiple columns selection
10.	How do you select rows by label (index)?	.loc[]
11.	How do you select rows by position?	.iloc[]
12.	How do you filter rows based on a condition?	Boolean indexing
13.	How do you add a new column to a DataFrame?	Assigning new column
14.	How do you drop a column?	drop() – columns
15.	How do you drop rows?	drop() – rows
16.	How do you handle missing values?	isnull(), fillna(), dropna()
17.	How do you find duplicate rows?	duplicated(), drop_duplicates()
18.	How do you group data and calculate aggregates?	groupby() and aggregation
19.	How do you merge two DataFrames?	merge()
20.	What is the difference between merge(), join() and concat()?	Combining DataFrames
21.	How do you sort a DataFrame?	sort_values(), sort_index()
22.	How do you rename columns?	rename()
23.	How do you reset the index?	reset_index()
24.	How do you apply a function to a column?	apply(), map(), applymap()
25.	How do you save a DataFrame to a CSV file?	to_csv()

Key Pandas Concepts

- Data Structures: Series, DataFrame, Index
- Data Selection: loc, iloc, filtering
- Data Cleaning: Missing values, duplicates
- Data Manipulation: Add/Drop, Rename, Sort
- Data Aggregation: groupby, aggregation
- Data Combination: merge, join, concat
- I/O Operations: read_csv, to_csv, read_excel



Tips for Interviews

- ✓ Understand the DataFrame structure deeply.
- ✓ Practice with real datasets.
- ✓ Know the differences between similar functions.
- ✓ Be ready to write small code snippets.
- ✓ Explain your thought process clearly.



Remember: Practice is the key! The more you work with Pandas, the more confident you'll be in interviews.